EX NO. 3 a DATA ANALYTICS ABOUT THE DATASET

AIM

To perform data analytics on a given dataset using Python and Pandas, understand the characteristics of the dataset, identify different types of data, examine missing values, and prepare the dataset for further analysis and machine learning.

INTRODUCTION

Data Analytics is the process of examining, cleaning, transforming, and interpreting data to discover useful information, patterns, and relationships.

A dataset usually contains:

- Rows – individual observations or records.
- Columns – attributes or features describing each record.
- Numerical data – values represented by numbers.
- Categorical data – values represented by categories or labels.

In this experiment, the dataset contains information about customers and their purchasing behavior. The four attributes are:

Country, Age, Salary, and Purchased.

The uploaded notebook shows that the dataset contains 10 records and 4 columns.

THEORY

1 What is Data Analytics?

Data Analytics is the systematic process of studying data to obtain meaningful information.

The general data analytics process is:

Data Collection → Data Understanding → Data Cleaning → Data Transformation → Data Analysis → Interpretation

For example, an e-commerce company may collect customer information such as:

- Country
- Age
- Salary
- Product purchased

By analyzing this information, the company can identify customer purchasing patterns.

DATASET

A dataset is a collection of related data organized in the form of rows and columns. It is used to store, analyze, and understand information. In this experiment, the dataset contains customer details such as Country, Age, Salary, and Purchased.

Publicly Available Dataset

- A publicly available dataset is a collection of data that is made accessible to students, researchers, developers, or the general public.
- It can usually be downloaded or accessed online for analysis and research.
- Public datasets are commonly used for:
    - Data Analytics
    - Data Science
    - Machine Learning
    - Artificial Intelligence
    - Research
    - Laboratory experiments
- Public datasets may be available in formats such as:
    - CSV
    - Excel
    - JSON
    - XML
    - Text
    - Images
    - Audio
- Examples of public dataset sources:
    - UCI Machine Learning Repository
    - Kaggle
    - Government Open Data Portals
    - Google BigQuery Public Datasets
- India's Open Government Data Platform (data.gov.in) provides publicly accessible datasets in areas such as education, health, environment, economy, and science and technology.
- Public datasets are useful because they allow students to practice analytics without collecting data themselves.
- Examples:
    - Iris Dataset – flower classification
    - Heart Disease Dataset – healthcare analysis
    - Student Performance Dataset – education analytics
    - Online Retail Dataset – customer purchasing analysis
    - Weather Dataset – environmental analysis

Real-Time Dataset

- A real-time dataset contains data that is generated or collected continuously from a real-world system.
- The data is made available for processing and analysis immediately or with very little delay after it is generated.

- Real-time data is often produced by:
  - Sensors
  - IoT devices
  - Mobile applications
  - GPS devices
  - Smart meters
  - Websites
  - Financial systems
  - Medical devices
- Real-time data can continuously change as new observations arrive.
- It is commonly used for real-time monitoring, prediction, and decision-making.

Examples

- Smart Traffic
  - Vehicle speed
  - Vehicle count
  - Traffic density
  - Location
- Smart Grid
  - Electricity consumption
  - Voltage
  - Current
  - Power usage
  - Smart-meter readings
- Healthcare
  - Heart rate
  - Blood pressure
  - Temperature
  - Oxygen level
- Weather
  - Temperature
  - Humidity
  - Rainfall
  - Wind speed
- Banking
  - Transactions
  - Payments
  - Account activity
- E-Commerce
  - Customer clicks
  - Product views
  - Orders
  - Purchases
- Transportation

- Vehicle location
- Speed
- Travel time
- Traffic conditions

Conclusion

Thus, the dataset and data analytics concepts were studied successfully, including publicly available and real-time datasets and their applications.